

motion tracking device that comprises a 3-axis gyroscope and a 3-axis accelerometer. The device enables high-precision motion sensing for applications such as robotics, wearable devices, drones, and many other IoT devices.



Compared to traditional consumer-grade IMUs, the ICM-42688-P provides a forty per cent lower noise-figure and improves temperature stability by 2x. This ensures that any motion event is measured with the highest accuracy and is independent of temperature changes.

The ICM-42688-P also includes a 100 per cent accurate clock and high-resolution analogue-to-digital converter. The converter facilitates an 8x increase in gyroscope resolution and a 4x increase in accelerometer resolution.

It is specifically designed with a focus on their tier-1 and channel customers, who care about high-precision and best-in-class sensors performance.

TDK
www.tdk.com

Notebook processor

AMD has announced the global availability of 7nm commercial notebook processors, the AMD Ryzen PRO 4000 Series Mobile family that delivers



the most cores and threads in an ultrathin business notebook. These new processors are fully optimised for remote work capabilities and designed to take business computing to the next level with multi-threading performance for modern productivity.

The AMD Ryzen PRO 4000 Series mobile processors are designed to deliver advanced computing and security. AMD Ryzen 7 PRO 4750U is the fastest business processor for ultra-thin business notebooks, with up to eight cores and sixteen threads and up to 33 per cent better multi-threaded performance compared to the competition. It delivers a responsive and power-efficient solution, with up to 37 per cent better overall performance in Microsoft Office applications over the previous generation.

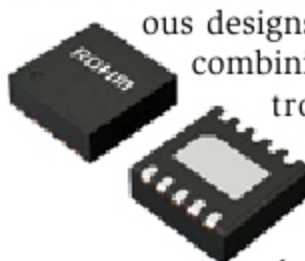
Built with untethered productivity in mind, along with the advanced 7nm

process and Zen 2 core architecture, the AMD Ryzen 7 PRO 4750U delivers up to 2x performance per watt over the previous generation.

AMD
www.amd.com

LED driver IC

Automotive lamps are increasingly adopting LEDs due to their longer life and high-density mounting. Conventional onboard LEDs support various designs and functions by



combining numerous electronic components on a modular substrate. Also, socket-type LED lamps have been

attracting attention by offering superior maintainability. However, it is difficult to decrease the socket size for better design flexibility.

In response, ROHM has developed a new, compact, and high output linear LED driver IC, BD18336NUF-M. It enables mounting in ultra-compact socket-type LED lamp circuits while ensuring stable lighting even during a drop in battery voltage. It is ideal for a broad range of socket-type LED lamps used in vehicle systems, from DRLs (Daytime Running Lamps) and position lighting to rear lamps.

The BD18336NUF-M has a current bypass function that prevents LED turn off. A smooth switching of the LED lighting path occurs through which current flows from three lamps to two without flickering when the supply voltage from the vehicle battery drops to 9V and maintains thirty per cent brightness at all times. This function can be configured using three external resistors (for the operating start voltage and operating interval), providing greater flexibility for a wide range of applications.

Rohm
www.rohm.com

POWER SUPPLIES

Power management ICs

Silicon Labs has announced a new line of energy-friendly power management ICs (PMICs) for EFR32 wireless devices and EFM32 microcontrollers (MCUs). The EFP01 PMIC family provides a

flexible and system-level power management solution enhancing the energy efficiency of battery-powered applications including IoT sensors, asset



tags, smart meters, and health and wellness products. These feature-rich PMICs enable developers to choose the optimal battery type and chemistries for their applications while controlling a product's power supply over multiple output rails and voltages.

Silicon Labs' PMIC solution addresses the power management needs of IoT developers by extending the energy efficiency of its wireless and MCU products while simplifying product design with best-in-class tools and support.

EFP01 PMICs simplify the power system design and reduce power consumption through enhanced control. The PMICs include low-voltage DC-DC converters and regulators, along with a flexible mechanism to manage the power rails in system design.

Silicon Labs
www.silabs.com

Wi-Fi SoC

Dialog Semiconductor, a leading provider of battery management, has announced the availability of



the DA16200, a highly-integrated and ultra-low-power Wi-Fi networking SoC to deliver a breakthrough in battery life for Wi-Fi connected battery-powered IoT devices.

Unlike other Wi-Fi SoCs, the DA16200 is specifically optimised to support such battery-powered IoT devices. DA16200's VirtualZero technology enables a low level of power consumption that makes it achieve three to five years of battery life. The SoC utilises an algorithm-driven design to provide the lowest power solution for facilitating an extended battery life, while also maintaining a continuous Wi-Fi connection.

To achieve extended range without compromising battery life, the DA16200 also features an integrated power amplifier (PA) and low-noise amplifier (LNA),